CS3001-1_Computer Network and Communication

Unit II MCQ

1. When two computers using IPv6 want to communicate but the packet must pass through an IPv4 region, which transition strategy should be used?
   1) Tunneling
   2) Header translation
   3) Fragmentation
   4) Either tunneling or header translation
   Ans: 1
2. When the load in the network reaches the network capacity, the packet delay _____ and the network throughput reaches _____.
   1) increases; its minimum
   2) increases; its maximum
   3) decreases; its minimum
   4) decreases; its maximum
   Ans: 2
3. The main header of an IPv6 protocol is _____ bytes.
   1) 20
   2) 40
   3) 60
   4) 80
   Ans: 2
4. A router is a connecting device that has functionality of the _____ layer(s).
   1) physical
   2) physical and data link
   3) data link and network
   4) physical, data link and network
   Ans: 4
5. Routing inside an autonomous system is referred to as _____ routing.
   1) intradomain
   2) interdomain
   3) out-of-domain
   4) None of the choices are correct
   Ans: 1
6. In mobile IP communication, the third phase is called _____ .
   1) Agent Discovery
   2) Registration
   3) Data Transfer
   4) Node Identification
   Ans: 3
7. Which QoS technique involves dropping or delaying packets to ensure that high-priority traffic is not adversely affected?
   1) Policing
   2) Shaping
   3) Queuing
   4) Congestion Management

Ans: 4

8. In the context of packet scheduling, what does "Fair Queuing" aim to achieve?
   1) Ensuring each flow receives an equal share of network resources
   2) Prioritizing packets based on their arrival time
   3) Maximizing throughput by merging multiple queues
   4) Minimizing the delay for high-priority packets
   Ans: 1
9. _____ allows a site to use a set of private addresses for internal communication and a set of global Internet addresses for communication with the rest of the world.
   1) DHCP
   2) NAT
   3) IMCP
   4) ISP
   Ans: 2
10. Which of the following QoS techniques is used to control the rate of traffic flow into the network to prevent congestion?
    1) Traffic Shaping
    2) Load Balancing
    3) Packet Switching
    4) Congestion Avoidance
    Ans: 1
11. In a datagram approach, the forwarding decision is based on the value of the _____ field in the packet header.
    1) Source Address
    2) Destination Address
    3) Label
    4) Cost of the link
    Ans: 2
12. In subnetting, the number of addresses in each subnet must _____.
    1) be a power of 2
    2) be a multiple of 128
    3) be divisible by 128
    4) None of the choices are correct
    Ans: 1
13. If the fragment offset has a value of 100, it means that _____.
    1) The datagram has not been fragmented
    2) The datagram is 100 bytes in size
    3) The first byte of the datagram is byte 800
    4) None of the choices are correct
    Ans: 3
14. The total length field defines the total length of the datagram _____.
    1) Including the header
    2) Excluding the header
    3) Excluding the option length
    4) Only Payload
    Ans: 1
15. Which of the following is true about ICMP messages?
    1) An ICMP error message may be generated for an ICMP error message.
    2) An ICMP error message may be generated for a fragmented datagram.
    3) An ICMP error message may be generated for a multicast datagram.

4) None of the choices are correct
Ans: 4

16. _____ is an interdomain routing protocol using path vector routing.
    1) BGP
    2) RIP
    3) OSPF
    4) Distance Vector
    Ans: 1
17. The _____ routing uses the Dijkstra algorithm to build a routing table.
    1) Distance vector
    2) Link state
    3) Path vector
    4) Link Cost
    Ans: 2
18. The _____ protocol allows the administrator to assign a cost, called the metric, to each route.
    1) BGP
    2) RIP
    3) OSPF
    4) All the choices are correct
    Ans: 3
19. In _____ routing, the least cost route between any two nodes is the route with the minimum distance.
    1) Path vector
    2) Distance vector
    3) Link state
    4) None of the choices are correct
    Ans: 2
20. The protocols _____ in version 4 are combined into one single protocol, as ICMPv6.
    1) ARP and IGMP
    2) ICMP and IGMP
    3) ICMP, ARP, and IGMP
    4) None of the choices are correct
    Ans: 3
21. What is the other name for distance vector routing algorithm?
    a) Ford and Fullman algorithm
    b) Bellman – Ford algorithm
    c) Ford and Fulkerson algorithm
    d) Bellman – Fulkerson algorithm
22. Which protocols are considered as intra-domain routing protocols?
    a) Distance vector and link state routing
    b) Distance vector and path vector routing
    c) Path vector and link state routing
    d) Only a path vector routing
23. What is flooding in link state routing in case of exchanging topology information?
    a) Exposing routing tables to neighbors
    b) Sending duplicate packets to nodes

c) Communicating with every router using messages

d) Spreading link state packets

24 Which sequence of procedure is correct to build a routing table in link state routing?

a) Creating LSP – flooding – shortest path tree – calculating routing table

b) Shortest path tree – flooding – creating LSP – calculating routing table

c) Creating LSP – flooding – calculating routing table – shortest path tree

d) Shortest path tree – flooding – calculating routing table -creating LSP

25 Which algorithm is implemented based on link state routing?

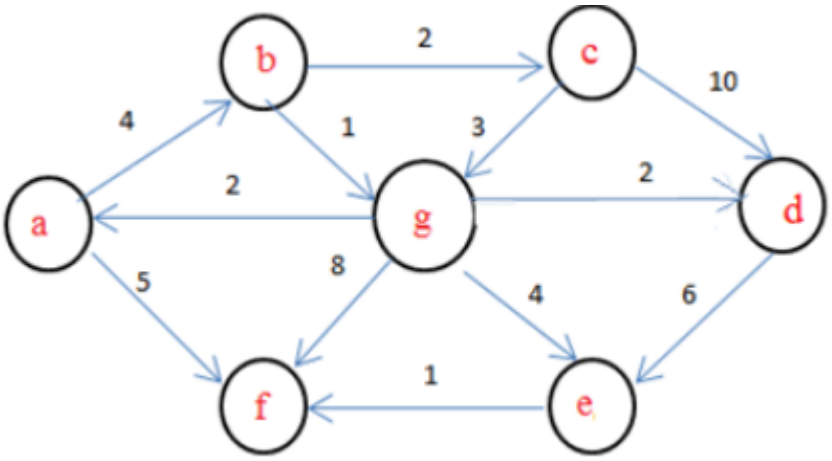
a) OSPF

b) BGP

c) RIP

d) OSLR

26 Consider the following graph.



If b is the source vertex, what is the minimum cost to reach f vertex?

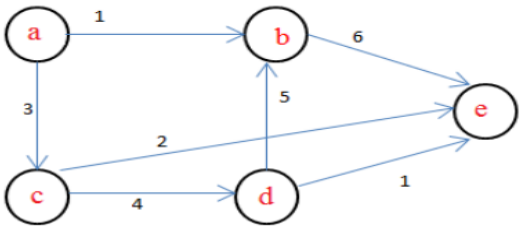
a) 8

b) 9

c) 4

d) 6

27 In the given graph, identify the shortest path having minimum cost to reach vertex E if A is the source vertex.



a) a-b-e

b) a-c-e

c) a-c-d-e

d) a-c-d-b-e

28 Which Class is reserved for future use?

a) A

b) B

c) D

d) None of the Mentioned

29 A classless address is given as 167.199.170.82/27. Find the number of addresses.

a) 128

b) 64

c) 32

d) 16

Explanation: n =27

Therefore number of addresses is $2^{(32-n)} = 32$.

30 CIDR stands for –

a) Class In Domain Range

b) Classless Inter Domain Routing

c) Classless In Domain Range

d) None of the mentioned.

31 Each IP packet must contain

a) Only Source address

b) Only Destination address

c) **Source and Destination address**

d) Source or Destination address

32 What is start frame delimeter (SFD) in ethernet frame?

a) 10101010

b) 10101011

c) 00000000

d) 11111111

33 MAC address is of _____

a) 24 bits

b) 36 bits

c) 42 bits

d) 48 bits

34 High speed ethernet works on _____

a) coaxial cable

b) twisted pair cable

c) optical fiber

d) unshielded twisted pair cable

35 Which class does the IP address 192.168.10.5/24 represent?

a) Class A

b) Class C

c) Class B

d) Class D

36 In IPv4 Addresses, classful addressing is replaced with _____

a) Classless Addressing

b) Classful Addressing

c) Classful Advertising

d) Classless Advertising

37 Which of this is not a class of IP address?

a) Class E

b) Class C

c) Class D

d) Class F

38 Which one of the following is not a function of network layer?
a) routing
b) inter-networking
c) congestion control
d) error control
Answer: d

39 A 4 byte IP address consists of _____
a) only network address
b) only host address
c) network address & host address
d) network address & MAC address

Answer: c

40 In virtual circuit network each packet contains _____
a) full source and destination address
b) a short VC number
c) only source address
d) only destination address
Answer: b

41 Which of the following is not correct in relation to multi-destination routing?
a) is same as broadcast routing
b) contains the list of all destinations
c) data is not sent by packets
d) there are multiple receivers

Answer: c

42 The network layer protocol for internet is _____
a) ethernet
b) internet protocol
c) hypertext transfer protocol
d) file transfer protocol
Answer: b

43 Which of the following field in IPv4 datagram is not related to fragmentation?
a) Flags
b) Offset
c) TOS
d) Identifier
Answer: c

44 The TTL field has value 10. How many routers (max) can process this datagram?
a) 11
b) 5
c) 10
d) 1

Answer: c

45 What should be the flag value to indicate the last fragment?
a) 0
b) 1
c) TTl value
d) Protocol field value

Answer: a

46 IPv6 does not use _____ type of address.
a) broadcast
b) multicast
c) anycast
d) unicast

Answer: a

47 Which among the following features is present in IPv6 but not in IPv4?
a) Fragmentation
b) Header checksum
c) Options
d) Anycast address
Answer: d

48 . The _____ field determines the lifetime of IPv6 datagram
a) Hop limit
b) TTL
c) Next header
d) Type of traffic

Answer: a

49 Header size of the ICMP message is _____
a) 8-bytes
b) 8-bits
c) 16-bytes
d) 16-bits

Answer: a

50 During error reporting, ICMP always reports error messages to _____
a) Destination
b) Source
c) Next router
d) Previous router

Answer: b

51 Which of these is not a type of error-reporting message?
a) Destination unreachable
b) Source quench
c) Router error
d) Time exceeded

Answer: c

52 ICMP error message will not be generated for a datagram having a special address such as _____
a) 127.0.0.0
b) 12.1.2
c) 11.1
d) 127

Answer: a

53 The source-quench message in ICMP was designed to add a kind of _____ to the IP.
a) error control
b) flow control
c) router control
d) switch control

Answer: b

54 Which of the following is not a similarity between flooding and distance vector routing?
a) Finding shortest paths
b) Using less setup
c) Using local knowledge
d) Using hop count metric

Answer: b

55 Which statement is correct about routing tables in distance vector routing?
a) Routing tables have entry for only immediate neighbouring nodes
b) Routing tables have entry for all the nodes in the network
c) Routing tables have entry for those nodes who have minimum distance
d) Routing tables have entry for those nodes who have maximum distance
Answer: b

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b) Shortest path tree – flooding – creating LSP – calculating routing table
c) Creating LSP – flooding – calculating routing table – shortest path tree
d) Shortest path tree – flooding – calculating routing table -creating LSP
Answer: a

57 A 4 byte IP address consists of _____
a) only network address
b) only host address
c) network address & host address
d) network address & MAC address
Answer: c